Fachhochschule Nordwestschweiz
Hochschule für Technik

Deep Learning

Studiengang Data Science

HS 2025

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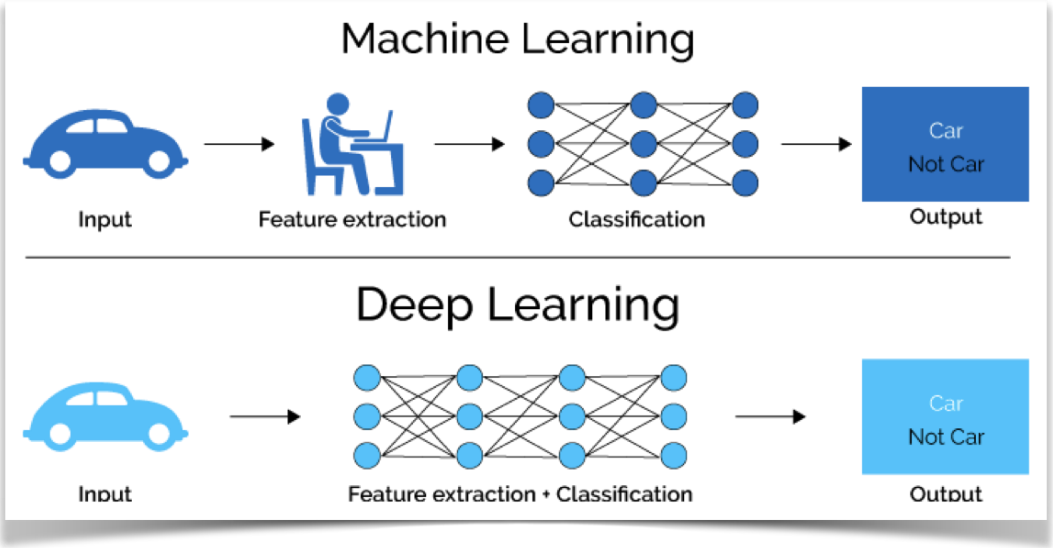
What is Deep Learning?

Machine Learning
-> Artificial Neural Nets (ANN)
    -> Shallow Neural Nets
    -> Deep Neural Nets (DNN)
        (many Layers)



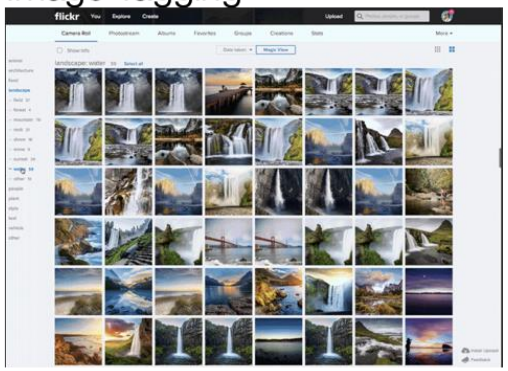
Kein Feature Engineering

Tasks, die traditionell einfach für Menschen, aber schwierig für Maschinen:

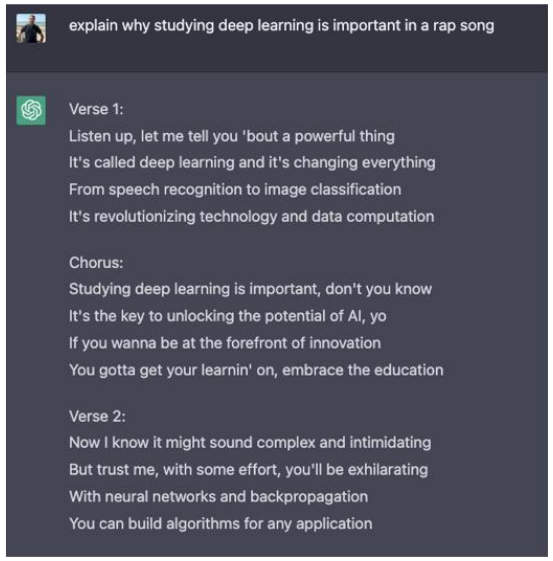
- Vision: Erkennen, Detektieren,
- Sprache: Verstehen, Übersetzen,
- Annotieren, Captioning

Lip reading, lip sync from audio

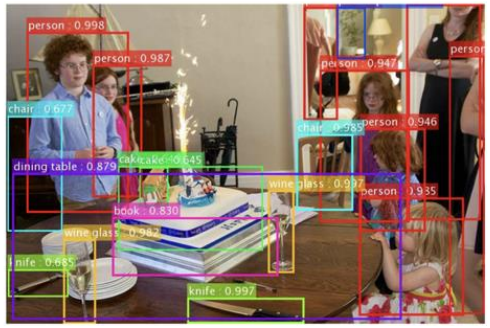
Image Tagging



Question / Answering

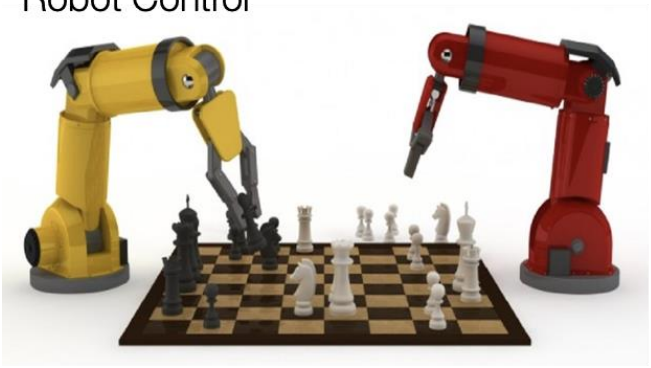


Object Recognition



Self-Driving Cars

Robot Control



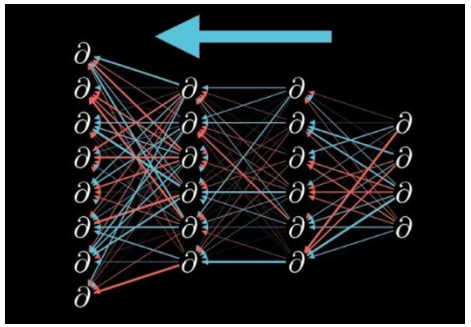
Text to image

Why Deep Learning Now?

Big Data

Larger quantities of data
text, audio, images, videos, ...

New algorithms
DBN, RBM, CNN ...



Deep Learning

Better computer performance
GPU, distributed computing ...

1990's
500 neurons
50K params
months of training on cpu

2010's
500'000 neurons
60M params
days/weeks of training on gpu and specifically designed hardware

2020's
up to 100 layers
up to 100 Billion (and more) parameters
days/weeks/months of gpu

See e.g. GPT3 with 175 Billion params
https://openai.com/
Switch Transformer 1.6 Trillion params
https://arxiv.org/pdf/2101.03961.pdf

5 Lernergebnisse

- LE1 - Multi-Layer Perceptron : 10%
- LE2 - Trainieren von Neuronalen Netzen : 40%
- LE3 - Convolutional Neural Networks (CNN) : 15%
- LE4 - Transfer Learning, Foundation Models: 15%
- LE5 - Deep Learning Frameworks: 20%

Organisation des Kompetenzerwerbs

- Auswahl der Lehrmittel
- Studium der Lerninhalte
- 1 Mini-Challenge, Praktische Übungen

} 5 Lernergebnisse

- Sprechstunde
  - Inputs
  - Klärung von Fragen, Unterstützung bei den Übungen
  - Di, Mi
- Deep Dive: 29 Sept 25, 09:15-11:00

Kurs Unterlagen

siehe space oder gitlab

- Coursera "Deep Learning Specialization" by Andrew Ng
- Stanford University "Convolutional Neural Networks for Visual Recognition" by Fei-Fei Li
- University of Geneva "Deep Learning Course" by Francois Fleuret

Mini-Challenge: Trainieren und Tunen von NN-Modellen

- Implementierung eines Deep Learning Modells (CNN, MLP) mit Hilfe eines DL-Frameworks (pytorch oder tensorflow)
- Anwendung auf einen Beispiel-Datensatz
- Modell-Selektion / Hyperparameter-Tuning
- Anwendung verschiedener Regularisierungsverfahren und Optimizers
- Abgabe in zwei Etappen:
  - Basismodell mit angepasster Lernrate und Batchsize; Feedback vom Fachexperten
  - Finale Abgabe mit Kurz-Präsentation und Q&A (~30min)

Termine (Vorschlag)

Date	Activity
16 Sep 2025	Kickoff: Org of the class; setup pytorch
29 Sep 2025	Deep Dive: Training with backprop, CNNs
7/8 Oct 2025	Keine Kontaktstunde
21 Oct 2025	MC Submission (first stage) – assessed, but not graded
28 Oct 2025	Input: Dropout, augmentation
4 Nov 2025	Input: Batchnorm
11 Nov 2025	Input: Optimizers (Adam, etc)
18 Nov 2025	Input: Weight Initialization
25 Nov 2025	Input: Transfer Learning, Imagenet Architectures
16/17 Dec 2025	Final MC Submission & Presentations